9/19/2026

BASICS OF PYTHON

Artificial Intelligence Lab

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Class: BSCS F24 B



Lab No 02

1. **Implement** a Python program that stores the names of 6 students in a list. The program must: (a) print the list, (b) print the first and last student using both positive and negative indexing, (c) add a new student at the end using `append()`, (d) insert a new student at position 2 using `insert()`, (e) remove one student by name using `remove()`, and (f) print the final list along with its length using `len()`.

```
Input:
["Ali", "Sara", "Bilal", "Ayesha", "Hamza", "Zain"]

Sample Output:
First student: Ali
Last student: Zain
Final list: ['Ali', 'Sara', 'Noor', 'Bilal', 'Hamza', 'Zain', 'Usman']
Total students: 7
```

Code:

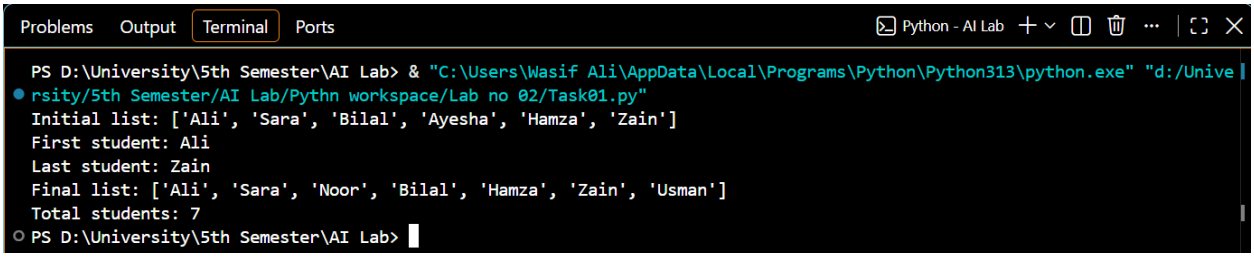
```python
students = ["Ali", "Sara", "Bilal", "Ayesha", "Hamza", "Zain"]

print("Initial list:", students)
print("First student:", students[0])
print("Last student:", students[-1])

students.append("Usman")
students.insert(2, "Noor")
students.remove("Ayesha")

print("Final list:", students)
print("Total students:", len(students))
```

Output:



2. **Apply** list creation with `range()` and a `for` loop to build a list of the first 15 natural numbers. Using a `for` loop (not `sort()`), separate them into two new lists: `even_numbers` and `odd_numbers`. Then use `sort()` on the original list in descending order, and use slicing to print only the middle five numbers of the sorted list.

```
Sample Output:
Numbers: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
Even numbers: [2, 4, 6, 8, 10, 12, 14]
Odd numbers: [1, 3, 5, 7, 9, 11, 13, 15]
Descending: [15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
Middle five: [10, 9, 8, 7, 6]
```

Code:

```python
numbers = []
for i in range(1, 16):
    numbers.append(i)

print("Numbers:", numbers)

even_numbers = []
odd_numbers = []

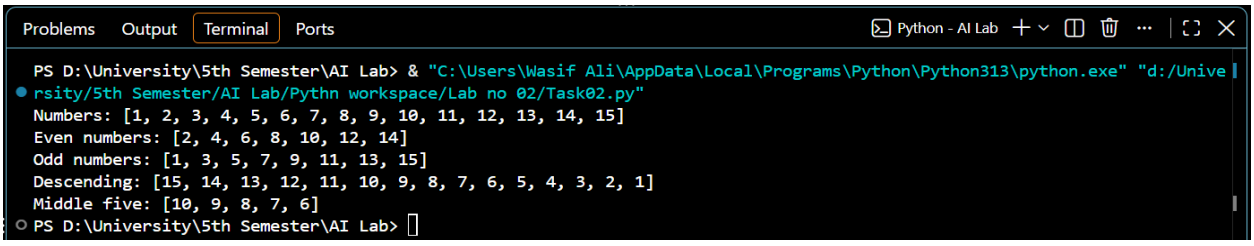
for num in numbers:
    if num % 2 == 0:
        even_numbers.append(num)
    else:
        odd_numbers.append(num)

print("Even numbers:", even_numbers)
print("Odd numbers:", odd_numbers)

numbers.sort(reverse=True)
print("Descending:", numbers)

middle_five = numbers[5:10]
print("Middle five:", middle_five)
```

Output:



3. **Implement** a Python program to manage a class attendance list, where each entry is either `"Present"` or `"Absent"` for 10 students. The program must: (a) display the full attendance list using a `for` loop, (b) use `len()` and a `for` loop to count and display the total number of students Present and Absent, (c) use slicing to display the attendance of only the first 5 and last 5 students, (d) reverse the list using `reverse()` and print it, and (e) calculate and display the attendance percentage (Present count / total students × 100).

```
Input:
["Present", "Present", "Absent", "Present", "Absent",
 "Present", "Present", "Absent", "Present", "Present"]

Sample Output:
Total Present: 7
Total Absent: 3
First 5: ['Present', 'Present', 'Absent', 'Present', 'Absent']
Last 5: ['Present', 'Present', 'Absent', 'Present', 'Present']
Attendance Percentage: 70.0%
```

Code:

```python
attendance = [
    "Present", "Present", "Absent", "Present", "Absent",
    "Present", "Present", "Absent", "Present", "Present"
]

print("Full Attendance:")
for hazri in attendance:
    print(hazri, end=" ")
print("\n")

total_students = len(attendance)
present_count = 0
absent_count = 0

for status in attendance:
    if hazri == "Present":
        present_count += 1
    elif hazri == "Absent":
        absent_count += 1

print("Total Present:", present_count)
print("Total Absent:", absent_count)

print("First 5:", attendance[:5])
print("Last 5:", attendance[-5:])
```

```python
attendance.reverse()
print("Reversed list:", attendance)

attendance_percentage = (present_count / total_students) * 100
print(f"Attendance Percentage: {attendance_percentage}%")
```

Output:

